Cockroach farming business plan

Eastern Asia between 39 and 40 degrees north latitude
Investment and operating strategy across all viable markets
Research cut-off 8 November 2026 | Financial currency USD

Recommended entry: validate a contained live-feeder business first. Develop pharmaceutical raw materials only with a qualified buyer and a defined legal route. Commit to industrial capacity only after paid trials, repeat orders and a full winter operating cycle.



Photograph P1. Blaptica dubia at Schwerin Zoo, Germany, 2007. Drägüs, CC BY-SA 3.0. An actual captive colony, not the proposed commercial facility. Full credits on page 28.

This plan compares live feeders, breeding stock, dried whole insects, meal and oil, pharmaceutical raw materials, research products, chitin and chitosan, frass and waste-treatment services. It sets out process design, investment envelopes, operating assumptions, break-even tests and staged expansion.

Decision status: suitable for pre-feasibility and buyer discussions. Location-specific permits, East Asian transaction prices, construction quotations and measured colony performance remain investment conditions. Financial scenarios are estimates, not supplier quotations or promised returns.

The investment decision

Start with customer demand and a small controlled trial

The strongest initial proposition is a dependable local supply of correctly sized, traceable live feeder insects for reptile keepers, specialist retailers and breeding facilities, where the chosen species and use are lawful. Repeat delivery, accurate counts and healthy arrivals are more defensible advantages than low-cost breeding stock alone.

A second route is contracted Periplaneta americana raw material for an established pharmaceutical manufacturer. It requires the buyer to approve the stock, feed, farm, harvest specification, processing method and analytical tests before expansion. The existence of large integrated Chinese producers does not establish that they will buy third-party insects. [S15]

Model	Illustrative funding	Base mature operating surplus	Judgment
Small live-feeder farm	$46,000	$300/month	A learning and local-sales operation; thin margin after owner labor.
Medium live-feeder farm	$290,000	$2,500/month	Only expand if price, recurring volume or cost improves.
Large live-feeder network	$2.21 million	$8,000/month	Weak base return; demand risk dominates.
Industrial dried raw material, 20 t fresh/month	$3.0 million	$31,500/month at assumed $15/kg dried	Conditional specialty-price case; loss at assumed $5/kg.

All figures are planning assumptions with explicit calculations on pages 17-22. Operating surplus means revenue less modeled cash operating costs, including labor, before depreciation, financing, taxes, replacement capital and working-capital changes. The industrial case is a separate plant, not an add-on to the live-feeder network.

Release capital in stages

Stage 1: spend $5,000-$10,000 on buyer validation, stock trials and metered winter testing. This is a limited experiment, not the fully funded small farm. **Stage 2:** install the small commercial unit after product and containment validation. **Stage 3:** replicate only when actual sales cover the next unit’s break-even volume. **Stage 4:** commission bulk processing against enforceable specifications and purchase commitments.

> Do not finance a new drug-development program, a speculative waste-disposal plant, and an insect farm as one startup. They have different buyers, operating controls and approval requirements.

Scope and evidence quality

The site is specified only as Eastern Asia at 39-40 N. Beijing and Akita illustrate two climates in this latitude band; neither is assumed to be the actual site. This band also crosses the Korean peninsula, where national jurisdiction matters. The plan uses USD to compare options without implying a current exchange rate.

Evidence type	What it can establish	How it is used
Primary experiments	Measured conditions, sample outcomes and limits	Species, feed and process design starting points.
Official rules and climate records	Published regulatory framework and station normals	Legal screening and cold-season engineering.
Company and facility reporting	Existence of operations and reported projects	Industry direction; not audited market size or profitability.
Public seller listings	Advertised asking prices for specific units	Breeding-stock and feeder benchmarks; not regional demand.
Planning assumptions	A transparent scenario for a proposed operation	Capital budgets, staffing, selling prices, yield and ramp.

Important limits on interpretation

No defensible, audited global cockroach-farming revenue series or country-by-country production census was established in this research. A broad “insect protein market” forecast would include other species and cannot be treated as the addressable cockroach market. Market sizing in this plan therefore starts with named customer categories and testable monthly orders.

Published counts may describe standing population, annual adults produced or total insects across multiple sites. These units are not interchangeable. A reported colony of billions does not reveal its saleable dry output, utilization, reject rate or return on capital.

Feed conversion also needs a stated basis: feed as supplied or feed dry matter, divided by live-weight gain or dry biomass. Breeders, mortalities, rejected batches and spilled feed may be absent from experimental ratios. Commercial management must include them.

Reading the plan

Pages 4-6 cover markets and legal screening. Pages 7-16 cover stock, feed, environment and operations. Pages 17-22 contain investment and financial scenarios. Pages 23-27 cover development, risks, selling and execution. Pages 28-31 provide photograph credits and source records. All citations link to the underlying source.

Global development and current patterns

Period or region	Observed evidence	Business implication
China, 2018	Xinhua reported 1 billion insects in a 6,300 m² waste-treatment farm consuming 50 t/day of kitchen waste. [S11]	Historical proof of a concept; no verified current intake, permitted feed use or margin.
China, 2025-2026	Reports describe digital breeding and integrated pharmaceutical processing. A 2025 article cites 25 extraction lines in a wider CNY 800m factory; a 2026 article reports 22bn insects. [S13] [S14]	Process control and pharmaceutical integration are significant. These figures cannot price a new farm.
North America	Specialist retailers publicly offer size-graded feeders and adult breeding groups. [S09] [S10]	An established niche distribution format; local East Asian pricing still needs evidence.
Europe	Industrial insect activity includes BSF and mealworms. 2026 reporting describes sector restructuring after Ynsect's late-2025 bankruptcy. [S30] [S31]	Scale and capital raised do not guarantee profitable production.
Research, 2024-2026	Recent studies test poultry substitution, poultry diets for Dubia and chitosan recovery. [S06] [S07] [S28]	New products remain dependent on economics, safety and intended-use authorization.

What is changing

The credible direction is toward controlled inputs, traceable batches, environmental sensors and closer links between farms and end users. Higher-value applications require consistent composition and analytical release, not just a large insect count. Heat integration and efficient logistics can matter more than elaborate automation at an early stage.

A 2018 report of six billion adults per year and later reports of much larger standing populations should not be used to calculate an industry growth rate: the definitions differ. [S12] [S14]

For this venture, the practical opportunity is a local service advantage or a contracted quality specification. A generalized "protein shortage" narrative is insufficient evidence to build capacity.

Compare the product markets

Product and customer	Potential attraction	Entry position
Live feeders: pet shops, reptile breeders, keepers	Recurring demand; sale by size and count; limited processing	First choice for pilot, subject to local legality.
Breeding stock: other insect keepers	Higher price per adult; supports initial cash receipts	Secondary product; avoid dependence on endless recruitment of new farmers.
Dried or frozen whole insects: specialty feed buyers	Simpler logistics than live shipments	Obtain intended-use clearance, processing validation and paid trials.
Meal and oil: aquaculture or livestock feed formulators	Potentially larger tonnage	Low priority until buyer price exceeds full production cost; compete with BSF, fishmeal and soy.
Medicinal raw material: licensed manufacturers	A documented Chinese application for P. americana	Contract-led route with buyer audit and species-specific quality specifications. [S19] [S26]
Chitin, chitosan and research extracts	Potential value from selected fractions or residues	Partner-led development; no base revenue assumed.
Research and teaching colonies	Small-volume specialist demand	Useful ancillary sales with traceable line identity.
Frass and waste-treatment services	Possible disposal saving or service fee	Separate permitted business case; zero revenue in the base plan.
Human food, supplements or cosmetics	Potential premium positioning	Do not launch without a confirmed legal route and safety dossier; zero base revenue.

Live feeders and processed ingredients need different unit economics. A retail price per insect includes grading, small-order handling and customer service. It cannot be multiplied by the number of insects in an industrial biomass shipment.

Pharmaceutical application is evidence of a specific commercial chain, not proof that any cockroach powder is a medicine, health food or safe cosmetic ingredient. No disease-treatment claims should be attached to farm products.

Regional legal and export screening

Before buying stock or signing a lease, request written determinations for the exact Latin species name, intended product, substrate, sales channel and proposed site. A live-feeder business, feed-ingredient plant and medicinal-material supplier may be classified differently.

Jurisdiction	Verified framework or restriction	Required project check
Mainland China	The 2025 feed-grain plan supports exploring insect protein; it is not species approval. The 2022 medicinal-material GAP framework includes animal-derived raw materials. [S18] [S19]	Ask agriculture authorities to confirm catalogue coverage for the exact insect and product; consult drug regulator and manufacturer for medicinal material.
Japan	MAFF feed-safety rules apply to regulated feed uses; JICA provides an insect-sector overview. [S20] [S33]	Confirm feed versus pet-feed classification, manufacturing notifications, import/quarantine, waste and local land use.
Republic of Korea	The MFDS 2024 edible-insect list contains ten categories and no cockroaches. [S21]	Confirm current MFDS food status, MAFRA feed rules, species possession/import and municipal requirements. This does not cover DPRK.
European Union export	The 2025 insect PAP species list contains flies, mealworms, crickets and silkworm; no cockroach species. [S22]	Do not include EU cockroach PAP sales in the base plan. Other products require separate classification.
United States export	Animal-food ingredients need a lawful intended-use basis. [S23]	Check federal ingredient status plus import, state species and feed requirements.
Australia export	Dubia and Madagascar hissing cockroaches cannot legally be possessed, traded or bred; permits cannot be issued. [S24]	Exclude live sales of these species. Do not infer permission for processed derivatives.

Also resolve building/fire approval, occupational exposure, water and wastewater, odor, transport, online selling, product labeling and end-of-life disposal. If the actual site is in the DPRK, rebuild the market, trade and legal assessment before applying the financial model.

Select and qualify breeding stock

Species	Commercial role	Operational judgment
Blaptica dubia	Live feeders and breeding groups	Preferred pilot candidate where lawful. Adult sexes are readily distinguishable. Maintain escape-proof containment even when adults climb smooth surfaces poorly.
Periplaneta americana	Medicinal raw materials; experimental feed and biomaterials	Use only for a contracted route. A recognized urban pest with higher consequence from escape; needs secure egg handling and facility containment. [S04] [S05]
Blaberus discoidalis and other feeder species	Possible regional feeder alternative	Assess species-specific law, buyer acceptance and husbandry separately; do not transfer Dubia yields automatically.
Gromphadorhina and display species	Research, teaching or pet trade	Niche product; not the baseline biomass organism. Check invasive-species restrictions.

Stock purchase specification

Buy captive-bred insects from identifiable suppliers with documented species, origin, generation or cohort, age range, sex counts, feed history, treatments and mortality history. Avoid wild-caught stock, pest-control collections and mixed unidentified colonies. Ask for actual reproductive records rather than a promise to buy all future offspring.

Obtain at least two independently maintained lines where lawful. Keep incoming lines physically separate for a proposed 30-day quarantine and longer if abnormalities appear. Inspect mites, fungal growth, abnormal deaths, egg or birth output and feeding. A veterinarian or insect-health laboratory should define testing when a pharmaceutical buyer requires it.

An incoming acceptance target for the pilot is at least 95% live arrival, correct species and sex count, and no unexplained mortality cluster. These are proposed contractual criteria, not established biological standards. Log every rejection and maintain a separate backup colony.

Starter stock budget benchmark

TopFlight lists 100 females plus 30 males at $200 and 200 females plus 60 males at $400. Telos lists 16 females plus four males at $20. These US retail examples imply roughly $1.25-$2.00 per breeding female with associated males, before delivery. Local availability and phytosanitary costs may differ substantially. [S09] [S10]

Photographic guide to Dubia stock



Photograph P2. Adult male, adult female and juvenile B. dubia, shown from top to bottom. Nicky.Animals, 2008; public domain. The scale of each panel is not standardized.

Use identity and cohort records together

The adult male’s long wings and the adult female’s short wing structures help stock sorting. Juveniles should be sold by measured size bands agreed with the customer. Do not use these photographs to infer insect weight, age or productive density.

Maintain separate breeding, nursery, grow-out and dispatch cohorts. Record active females, adult males, births, deaths, retained replacements and saleable output. A colony that appears crowded can still have low reproductive performance if its age distribution or sex ratio is wrong.

Start with a proposed male:female ratio around 1:3 to 1:4 and adjust from reproductive results; the public breeding packs above use similar ratios. Supplier packaging is a benchmark, not evidence of a universal optimum. [S09] [S10]

American cockroach identification and eggs



Photograph P3. Periplaneta americana, photographed in San Jose, California, 2016. Eugene Zelenko, CC BY-SA 4.0. Identification reference only; not approved breeding stock.



Photograph P4. P. americana egg capsule or ootheca. Eran Finkle, 2008, CC BY-SA 3.0. The capsule can carry viable offspring and must be included in waste and escape controls.

A controlled study at 29 C and 70 +/- 5% relative humidity reported about 35 days for embryonic development, 12-16 eggs per ootheca, and a mean four-day interval between oothecae in mated females. These are laboratory observations, not a promise of farm output. [S04]

Collect and date egg cases in secure incubation units. Monitor hatch rate per cohort, not just total egg cases. Do not treat unhatched capsules or old shelter material as inert waste. A validated kill step must cover eggs as well as mobile stages.

Production biology and capacity planning

Published observation	Interpretation for planning
Dubia diet trial at 28 C and 70% RH: development 200-294 days; survival 47-80%; fresh-basis FCR 1.5-2.7 across diets. [S01]	Do not assume a new colony reaches full breeder production in weeks. Small feeder grades may sell before adulthood.
Dubia thermal study: adulthood achieved at 20, 25 and 30 C; all nymphs died before fifth instar at constant 35 C and second instar at 40 C. [S03]	Use conservative operating bands, multiple sensors and over-temperature protection. Brief thermal tolerance is not a rearing setpoint.
P. americana at 29 C: about 35 days to hatch; reproduction differs between mated and unmated females. [S04]	Model hatchery, grow-out and breeder replacement separately.
Dubia composition varied with diet: dry matter 28-39%; crude protein 38-73% of dry matter. [S01] [S02]	Buyers need batch analysis. Crude protein is not identical to digestible true protein.

Transparent Dubia breeder model

Assumed saleable insects per active female per month = 15 newborns × 85% survival to sale grade × 90% available after replacement retention × 95% grade/order fulfillment = **10.90**. These factors are unvalidated commercial assumptions; no measured 85% farm survival is claimed.

At 2,000 / 20,000 / 200,000 active females, calculated capacity is about 21,800 / 218,000 / 2.18m saleable insects per month. Financial sales are capped at 20,000 / 200,000 / 2.0m. The replacement retention is included; maintain a separate off-room backup within the commissioning reserve.

If newborn output is only ten per female and survival falls to 70%, the same formula yields 5.99 saleable insects per female, a 45% reduction. Building more racks cannot fix an unmeasured reproductive bottleneck.

Track breeder-months, hatch/birth dates, time to each sale size, live mass per 100 insects, density per usable shelter area, feed intake and mortality. Farm capacity is the lowest of breeding, grow-out, sorting, packing and paying customer demand.

Feed specification and procurement

Use consistent approved inputs

Start with a documented, non-medicated commercial ration that is lawful for the intended insect/product chain. Obtain specifications and retain samples. A controlled comparison found rat chow better than beef or vegetable-only diets for P. americana reproduction and nymphal development. A 2026 Dubia study found similar growth and mortality across four poultry rations, while mineral balance remained a concern. [S08] [S06]

Illustrative 100 kg dry blend	Quantity	Assumed crude protein	Contribution
Ground maize	45 kg	8.5%	3.83 kg
Wheat bran	25 kg	15%	3.75 kg
Soybean meal	20 kg	46%	9.20 kg
Dried brewer’s yeast	10 kg	45%	4.50 kg
Total	100 kg	Calculated 21.3% as fed	21.28 kg

This is a trial formulation, not a validated optimum or pharmaceutical specification. Ingredient protein values are illustrative purchase assumptions. Analyze the final blend; a feed specialist should set mineral/vitamin additions at the supplier’s labeled rate and rebalance the recipe. Replace soy or yeast only through controlled comparisons.

Control the inexpensive-input trap

Specify moisture, protein, particle size, mycotoxins, pesticide residues, heavy metals and microbiological quality according to the product’s hazard assessment. Receive sealed lots with supplier and batch identifiers. Use first-expiring stock first; isolate moldy, wet, rancid or untraceable feed. Keep feed storage apart from frass and live dispatch.

Use potable water through clean dispensers or an approved hydration system. Offer moist feed in amounts consumed promptly and remove spoiled material. Measure any vegetable supplement separately: its high water content can distort the apparent feed conversion ratio.

Do not base feed-grade or medicinal production on sewage, manure, municipal mixed waste, uncontrolled restaurant scraps or pesticided produce. Substrate and processing determine contaminant risks, and heating does not remove all chemicals. [S25]

Feed trials and mass accounting

A practical pilot design

Compare three lawful feed options: a commercial control, the proposed plant-based blend and one locally available, traceable food-manufacturing by-product blend. Use six independent containers per treatment if resources permit; randomize container positions and keep initial age and density comparable. A container is the replicate, not each insect inside it.

Run growth measurements for 12 weeks as an early screen and continue the best treatments through reproduction and a winter period. Record feed offered and remaining on a dry-matter basis, live-weight gain, deaths, time to sale size, births, labor and electricity. Prefer the lowest full cost per accepted saleable insect, not simply the lowest feed price.

Measure	Formula or recording rule
Commercial feed conversion	All feed issued less recoverable residual feed, divided by accepted live biomass produced; include breeder feed and rejects consistently.
Dry-matter conversion	Feed mass × feed dry-matter fraction divided by insect gain × insect dry-matter fraction.
Feed cost per sold live kg	Measured whole-farm feed ratio × delivered feed cost per kg.
Net productive density	Accepted insects or kg per month divided by occupied productive room area; separately record shelter area and tiers.
Effective processing yield	Accepted packaged dry product divided by harvested fresh mass; record moisture in both.

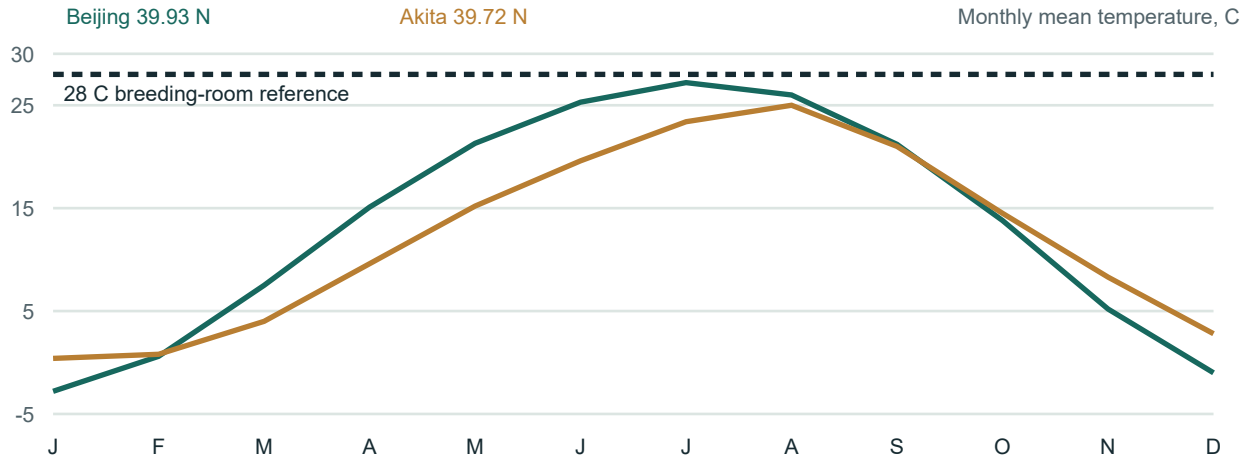
Illustrative dry-product balance

For **1,000 kg fresh insects** at 30% dry matter, assume 90% dry-solids recovery and finished product at 10% moisture. Saleable dried mass = 1,000 × 0.30 × 0.90 ÷ 0.90 = **300 kg**. The result contains 270 kg solids and 30 kg water. Approximately 670 kg water is removed and 30 kg dry solids leave as process loss, assuming no other water additions or losses.

At a whole-farm feed assumption of 2.5 kg as-fed ration per fresh kg and $0.35/kg feed, feed alone costs $875 per tonne fresh, or $2.92 per kg dried. Add labor, climate control, drying, testing, packaging, rent and financing before setting a sale price.

> Never add revenue from whole dried insects, extracted oil, purified protein and chitin as though each were produced from the entire same kilogram. Each fraction must reconcile to one mass balance.

Winter climate and energy strategy



Beijing's January normal is -2.8 C and July 27.2 C; Akita's January normal is 0.4 C and August 25.0 C. Both are within the requested latitude band, but coastal humidity and seasonal conditions differ. These averages are not minimum design temperatures. [S16] [S17]

Build insulated rooms inside a serviced building

Heat a compact, insulated breeding envelope rather than an entire warehouse. Specify vapor control, washable surfaces, thermal bridges, filtered ventilation, condensation drainage, calibrated room and bin sensors, independent high-temperature cut-outs and alarm power backup. Separate cohorts so one equipment failure does not affect every breeder.

Illustrative 100 m² room	Design assumption	Calculated result
Indoor/outdoor temperature	28 C / -10 C	38 C difference
Envelope heat loss	U = 0.30 W/m²K; area 340 m²	3.88 kW
Ventilation heat loss	Volume 300 m³; 0.5 air changes/hour	1.88 kW
Total sensible load	No heat recovery or insect heat credited	5.76 kW
30-day heating energy	5.757 × 24 × 30	4,145 kWh thermal
Purchased electricity	Heat-pump COP 2.5; $0.12/kWh	1,658 kWh; about $199/month

This engineering example excludes latent moisture loads, doors, fan power, defrost, dryer demand and extreme weather. At U = 1.0, 1 air change/hour and resistance heating, the same example reaches about 12,011 kWh and $1,441/month. Verify all inputs through a local engineer and a measured winter trial; do not treat either number as a site utility quote.

Facility layout and industrial flow

One-way product flow
Approved feed receipt → quarantine and breeding → nursery → grow-out → harvest and grading → live dispatch OR validated killing and drying → testing and release → finished-goods dispatch.

Separate dirty flow
Frass, mortalities, egg cases and used shelters → sealed collection → validated inactivation → permitted treatment or disposal.

Essential rooms and boundaries

Place stock quarantine and a backup breeding line in separate rooms. Use a two-door entry vestibule, sealed service penetrations, insect-proof vent screens, secure drain barriers and washable lower wall surfaces. Design barriers for the smallest mobile stage; examine damaged screen edges and door seals. Provide dedicated clothing and tools by zone.

Feed mixing, frass handling and dry milling should have local dust extraction and suitable filtration. Keep clean packaging away from live-insect dust and untreated waste. Use dedicated cleaning equipment; never push potentially viable insects or eggs into drains. Measure ventilation for humidity and worker exposure rather than reducing airflow simply to save heat.

Control and automation by stage

Small	Medium	Large
Lidded bins, hand sorting, weighing, digital batch log, two independent temperature alarms.	Modular insulated rooms, standardized racks, batch weighing, feeder carts, calibrated sizing screens, barcode lots.	Independent production modules, central monitoring, automated feed delivery where trials justify it, validated continuous or batch processing.
Owner checks every room daily and keeps a backup heat source.	Named production supervisor and release authority; planned maintenance.	Engineering redundancy, qualified quality manager, traceability system, spare parts and contracted emergency support.

Automation purchases should follow time-and-motion measurements. Handling injury, size variation and clogging can negate labor savings. Require a demonstration on the actual species and product size before paying for equipment.

The layout above is a proposed industrial process, not a photograph or certified engineering drawing. Have fire, ventilation, wastewater and biological containment specialists review the final building plan.

Daily operations and batch controls

Stage	Required activity	Release or stop rule
Incoming stock	Check identity, counts, condition and supplier documents; isolate new lots.	No mixing until quarantine acceptance.
Breeding	Record active females, sex ratio, feed, births/oothecae, deaths and sensor readings.	Investigate a sustained 20% output decline against the room's own baseline.
Nursery and grow-out	Use dated cohorts; measure size and mass samples; avoid wet residues and crowding.	Hold harvest after unexplained mortality or chemical exposure.
Harvest	Protect replacement stock; grade gently and reconcile quantities.	Reject contaminated lots; do not salvage unexplained deaths as feed.
Live packing	Count-check, ventilate, protect against heat/cold, apply batch identity and care information.	Ship only inside the validated route and weather window.
Dry processing	Use defined harvest age, validated lethality, drying endpoint and hygienic packing.	Quarantine finished lot until required laboratory tests pass.
Cleaning and waste	Remove mortalities daily; contain frass and egg cases; inactivate before exit.	No untreated viable material leaves the containment boundary.

Proposed operating bands

Begin Dubia trials at approximately 27-30 C and 55-70% room RH, with clean water and good airflow. P. americana trials can begin near the published 29 C and 70% RH conditions if approved by the buyer. These are starting bands requiring site validation, not universal standards. Map temperature inside loaded bins: room air can conceal hot or wet pockets. [S03] [S04]

Set early temperature warnings around 25 C and 31 C for the Dubia pilot and use independent escalation alarms. Investigate alarms immediately; the exact emergency thresholds and backup capacity must be determined by trial and engineering review. Include power loss, sensor failure, high humidity and water leakage.

At every shift change, record open deviations and restricted lots. Weekly review should reconcile insects sold, production, mortality, retained breeders, feed inventory and energy. Monthly review should compare actual cost and realized price to break-even.

Quality safety and worker protection

Cockroach rearing requires control of allergens, contamination and escapes even when the colony is captive-bred. Feed origin, insect stage and processing affect the hazard profile. Use a documented hazard analysis and qualified laboratory methods. [S25] [S32]

Hazard	Control	Verification
Microbial contamination	Approved feed and water; clean zoning; validated lethality for processed goods.	Risk-based raw and finished-product sampling; organism-specific buyer limits.
Pesticides, mycotoxins, metals	Qualified inputs; exclude treated/waste materials; supplier specifications.	Incoming and periodic product assays, adjusted after any input change.
Allergen and dust exposure	Enclosed frass/milling tasks, extraction, wet cleaning or HEPA vacuuming, protective clothing.	Occupational health assessment, training and appropriately fitted respiratory protection where needed.
Egg or insect escape	Nested containers, secure doors/screens/drains, perimeter monitoring.	Daily checks, trap map and documented incident drills.
Spoilage after drying	Dry to validated water activity, protect from moisture and oxygen.	Suggested development target $a_w \leq 0.60$; verify shelf-life and legal/buyer limits.
Batch misidentification	Unique feed-to-customer lot codes and controlled labels.	Mock recall tracing one lot forward and back within a proposed four hours.

Freezing immobilizes or kills under defined conditions, but it is not by itself a validated microbial decontamination step. Drying temperature alone is also insufficient evidence: validate time, core temperature, load depth, cold spots and survival of relevant organisms. Do not assume cooking eliminates allergens or chemical contaminants.

Batch specification for a processed product

Agree species identity, permitted substrate, harvest stage, moisture, water activity, crude protein method, fat, ash, microbial criteria, chemical contaminant limits, particle size, storage conditions, packaging and shelf life with the buyer. Medicinal buyers also need their analytical fingerprint or marker specification and supplier qualification.

Label development targets as provisional. For example, 10% moisture in the financial model is an assumed sale basis, not a universal safe-storage specification. If the tested water activity or buyer standard requires lower moisture, recalculate yield and drying cost.

Scale definitions and production resources

Live-feeder model	Small	Medium	Large
Paid sales per month	20,000 insects	200,000 insects	2,000,000 insects
Annual sales at maturity	240,000	2.4 million	24 million
Active female target	2,000	20,000	200,000
Male planning range	500-670	5,000-6,700	50,000-67,000
Total premises allowance	30-60 m²	200-400 m²	1,500-3,000 m²
Staffing allowance	Owner, part time	2-4 equivalent full-time roles	12-20 equivalent full-time roles
Physical output at 0.30 g average	6 kg live/month	60 kg live/month	600 kg live/month
Daily packing at 22 dispatch days	909 insects	9,091 insects	90,909 insects

These are design allowances, not measured stocking densities or regional wage quotations. Floor area includes aisles, quarantine, feed and packing. The average 0.30 g sale weight is an assumption for a mixed feeder-size portfolio; weigh local samples and revise it. A “large” feeder network is large by counted sales, not by commodity biomass tonnage.

Production inventory and capacity acceptance

Allow four to eight months of grow-out capacity until actual size-grade lead times are known. At the large sales target, six months of sale-age pipeline implies roughly 12m future saleable insects before breeder and mortality allowances. This is a count-based planning proxy, not an estimate of standing live biomass.

Have the equipment supplier demonstrate feed access, room uniformity, cleaning time and yield at the proposed loaded density. Limit each room’s share of breeder capacity so one loss does not erase the business. Retain spare bins and floor space for quarantine or abnormal cohorts.

Separate industrial biomass branch

For dried raw materials, the comparison is 250 / 2,000 / 20,000 kg fresh harvest per month, producing 75 / 600 / 6,000 kg dried product. These larger biomass requirements need their own breeding study, facility plan and budget. Do not infer them from the Dubia count-based sales model. See page 22.

Capital and working capital

All figures USD thousands. Concept-level allowances for leased, serviced premises; local quotations are required. No land purchase, new shell building, finished-drug factory or R&D approval program is included.

Use of funds	Small	Medium	Large
Insulation, fit-out and containment	4	40	350
Racks, bins and handling	3	25	250
Climate control, alarms and backup	3	30	200
Packing, scales and monitoring	2	15	100
Equipment and fit-out subtotal	12	110	900
Breeding stock and commissioning reserve	4	40	400
Pre-opening, design, testing and permits	2	15	100
Capital contingency	3	25	210
Opening working-capital cash	25	100	600
Total funding	46	290	2,210

Stock allowances use the upper US retail benchmark of roughly $2 per female with males as a conservative placeholder. They are not East Asian quotes; phased breeding or local sourcing can reduce purchases while lengthening ramp-up. Commissioning reserves cover initial replacements, but long disruptions need additional funds. [S09]

Investment ranges and spending sequence

Use broad total-funding envelopes of **$30,000-$70,000 small**, **$200,000-$450,000 medium** and **$1.5m-$3.5m large** for initial screening. The point estimates above are the model cases. Uncertain regulatory requirements or a new building can exceed these ranges.

Release payments against design approval, delivery, installation and performance acceptance. Negotiate spare parts, warranty, service response and power-loss restart. Keep working-capital cash liquid instead of spending it on larger equipment before sales materialize.

Obtain separate quotations for freight, installation, taxes, winter design, utility connections, waste handling and mandatory safety upgrades. If a buyer requires pharmaceutical-quality origin processing or additional laboratory capacity, add it to both capital and recurring cost.

Live-feeder operating economics

Monthly figures at full planned sales. Prices are hypothetical local net product revenue after discounts and refunds, before modeled fulfillment costs. Taxes and customer-paid freight are excluded. US prices on page 7 demonstrate a sales format, not local price validation.

Input or result	Small	Medium	Large
Sales count	20,000	200,000	2,000,000
Net price per insect	$0.140	$0.080	$0.045
Variable cost per insect	$0.030	$0.025	$0.020
Revenue	$2,800	$16,000	$90,000
Variable costs	$600	$5,000	$40,000
Fixed cash operating costs	$1,900	$8,500	$42,000
Operating surplus	$300	$2,500	$8,000
Operating margin	10.7%	15.6%	8.9%
Break-even sold count	17,273	154,546	1,680,000
Break-even net price	$0.125	$0.0675	$0.041

Fixed costs per month	Small	Medium	Large
Labor including owner compensation	$1,000	$4,500	$24,000
Rent	$300	$1,200	$6,000
Energy and water	$200	$1,000	$5,000
Quality, maintenance, insurance, admin and marketing	$400	$1,800	$7,000
Total	$1,900	$8,500	$42,000

Variable allowances combine feed/hydration/shelters ($0.004/$0.003/$0.002), packaging ($0.012/$0.010/$0.008), seller-paid delivery support ($0.007/$0.006/$0.005) and fees/consumables ($0.007/$0.006/$0.005). Base staffing is in fixed labor. These per-unit allowances must be replaced by invoices and measured labor minutes.

Break-even count = fixed costs ÷ (net price − variable cost). Break-even price = variable cost + fixed costs ÷ sold count. The low margin leaves little room for an unplanned wage, freight or heating increase.

Sensitivity and investment hurdles

Monthly operating surplus	Small	Medium	Large
Base case	$300	$2,500	$8,000
Price 20% lower; volume unchanged	-$260	-$700	-$10,000
Sales 30% lower; same unit margin	-$360	-$800	-$7,000
Energy cost doubles	$100	$1,500	$3,000
Price -20% and sales -30%	-$752	-$3,040	-$19,600
Premium price: $0.18 / $0.10 / $0.055	$1,100	$6,500	$28,000

Lower-volume cases reduce variable costs with sales and hold fixed costs constant. In reality, unsold live insects continue consuming feed and occupy space, so these cases can understate losses. The energy shock doubles only the energy-and-water line; a poorly insulated facility can face a larger increase.

Avoid confusing operating surplus with return on investment

Annualized mature base surpluses are $3,600 / $30,000 / $96,000. Total funding divided by those surpluses gives crude steady-state multiples of 12.8 / 9.7 / 23.0 years. These are not discounted payback periods and exclude ramp losses, replacement equipment, interest and tax. They show that the base cases are financially weak.

At the premium-price assumptions, annual operating surpluses become $13,200 / $78,000 / $336,000. The corresponding simple funding multiples are 3.5 / 3.7 / 6.6 years before ramp effects. These cases require paying customers at those prices; they are not forecasts.

Proposed expansion hurdle

For the medium farm, target an evidenced net price of at least $0.10 per insect at 200,000 monthly paid sales, or an equivalent verified cost improvement. At that price the modeled break-even volume falls to 113,334 insects/month. Pre-sell at least 70% of planned output to a diversified customer base before major fit-out.

Use equity or patient capital for biological validation. Add debt only after stable operating cash and realistic repayment coverage are demonstrated. Model borrowing rates and repayment schedules using actual financing offers; this plan assumes no debt and claims no IRR.

Ramp up and five year cash view

The deliberately cautious sales ramp is 20% of full annual sales in Year 1, 70% in Year 2 and 100% in Years 3-5. All fixed costs are incurred from opening; variable costs follow sales. Pre-opening spending is separate. No growth beyond installed capacity, grants, tax relief, frass income or stock revaluation is counted.

Annual operating surplus, USD thousands	Year 1	Year 2	Year 3	Year 4	Year 5
Small	-17.52	-4.32	3.60	3.60	3.60
Medium	-75.60	-9.60	30.00	30.00	30.00
Large	-384.00	-84.00	96.00	96.00	96.00

Cash check, USD thousands	Small	Medium	Large
Opening working-capital cash	25.00	100.00	600.00
Cumulative operating loss through Year 2	21.84	85.20	468.00
Cash left after Year 2	3.16	14.80	132.00
Cumulative operating surplus through Year 5	-11.04	4.80	-180.00

This cash view assumes immediate collection, immediate expense payment and no additional inventory buildup, capital replacement, taxes or financing. It is not a full accounting cash-flow statement. Breeding stock is funded upfront; routine replacement retention is embedded in productive output. Slower maturation, customer credit or an outbreak can require more cash.

Working capital policies

Use prepayment for consumer orders and short credit limits for business accounts. At base monthly revenue, 30-day receivables would tie up another $2,800 / $16,000 / $90,000. A two-week feed stock, packaging inventory and retained test samples also require funding. Refresh a rolling 13-week cash forecast every week.

Set a cash floor equal to at least three months of fixed operating costs plus committed liabilities. The small and medium base cases fall below that floor during ramp, so lower spending, extra reserves or better price realization must be established before funding. Do not assume the opening reserve alone makes the plan financeable.

> The five-year base case does not recover the initial investment. Scale-up approval should depend on pilot-verified unit economics, not on the optimistic effect of more insects.

Dried raw material industrial case

This is a separate P. americana or buyer-approved biomass concept. Fresh harvest of 250 / 2,000 / 20,000 kg per month yields 75 / 600 / 6,000 kg finished dried material using page 12's 30% mass yield. The feed ratio of 2.5 is a scenario assumption, not a demonstrated industrial P. americana result.

Monthly input or cost	Small	Medium	Large
Fresh harvest, kg	250	2,000	20,000
Feed at 2.5 kg/kg, kg	625	5,000	50,000
Feed at $0.35/kg	$219	$1,750	$17,500
Labor	$2,000	$6,000	$18,000
Energy and water	$500	$1,500	$6,000
Packaging at $0.50/kg dried	$38	$300	$3,000
Rent	$300	$1,000	$5,000
Quality testing	$200	$1,000	$3,000
Maintenance and administration	$300	$1,500	$6,000
Total cash operating costs	$3,556	$13,050	$58,500
Break-even dried price per kg	$47.42	$21.75	$9.75

At an assumed specialty dried-raw-material price of **$15/kg**, monthly operating results are about **-$2,431 / -$4,050 / +$31,500**. At a hypothetical commodity price of **$5/kg**, they are **-$3,181 / -$10,050 / -$28,500**. Neither price is a verified regional quotation. No extract, oil or frass revenue is added.

For the large case, 60% utilization gives 3.6 t dried/month. Treat feed, packaging and half the energy as variable: costs fall to $49,100, revenue at $15/kg is $54,000, and operating surplus is only $4,900. At full volume but 25% fresh-to-dried yield, costs become $58,000 and break-even rises to $11.60/kg.

Illustrative total funding: **$60,000 / $400,000 / $3.0m**, comprising equipment/fit-out $35k/$240k/$1.9m, stock/pre-opening $5k/$30k/$200k, contingency $5k/$40k/$300k and cash reserves $15k/$90k/$600k. Plan roughly 40-80 / 300-600 / 3,000-6,000 m² subject to measured yield. These are order-of-magnitude budgets, not quotations. A finished-drug or solvent-extraction plant is excluded.

Develop products in an achievable order

First establish a reliable live product

Launch two or three size grades and a standardized adult breeding pack. Define minimum count, size tolerance, dispatch days and live-arrival policy. Retain cohort and feed records. A feeder is an ingredient in an animal's diet, not automatically a complete balanced diet; the 2026 Dubia study reported calcium:phosphorus ratios below 1:1. [S06]

Then qualify stabilized whole insects

Ask a specific buyer to approve a frozen or dried pilot lot. Compare freezing, thermal treatment and drying for product quality, microbial reduction, yield and cost. Begin with a qualified contract processor when feasible. Confirm that its handling of other animal materials does not conflict with the buyer's requirements.

Develop meal or oil only with a formulation customer

Measure amino-acid composition and digestibility, fat stability, chitin, ash, palatability and contaminants. A 2024 experiment used 350 Ross 308 broilers in seven treatments and reported favorable outcomes at 12% soybean-meal replacement with P. americana or BSF. That is one controlled study, not blanket feed approval or a commercial inclusion recommendation. [S07]

A September 2026 pig study further distinguishes full-fat from unfatted P. americana material: the full-fat diet had poorer protein and energy utilization; a diet containing 3% unfatted material in place of equivalent soybean protein improved finishing-pig daily gain. This supports buyer-led processing trials, not automatic equivalence of all cockroach meals or permission to sell them. [S34]

Medicinal raw materials require buyer-defined controls

Kangfuxin and Xinmailong demonstrate product-specific use of P. americana extracts in China. The proposed farm should supply qualified raw material under a quality agreement, not manufacture an unapproved therapeutic product. Define authorized harvest stage, feed, drying, marker or fingerprint tests, contaminants and change-control obligations. [S19] [S26]

Clinical research should not be converted into farm marketing claims. A 2025 Kangfuxin review included 19 studies but identified evidence-quality limitations. It cannot establish the efficacy of a newly made crude extract. [S27]

Keep specialty fractions as funded development projects

For chitosan, oil, hydrolysates or cosmetics, obtain a customer specification, independent safety review, regulatory classification and mass balance before buying processing equipment. Intellectual-property freedom to operate is part of that work. Assign zero revenue to unqualified products in the farm plan.

Processing routes and technical gates

Route	Industrial sequence	Decision gate
Whole dried material	Identity and sorting → validated killing → drying → cooling in clean air → moisture-barrier packing → lot release	Buyer-accepted chemistry, safety, stability and yield at three consecutive production lots.
Feed meal	Validated kill/dry → controlled milling → sieving → optional blending → metal control → packing	Legal ingredient status; mill throughput; particle-size and feed-trial acceptance.
Oil and protein fractions	Conditioning → validated separation → fraction stabilization → testing	Actual recovery and selling prices must cover extra energy, equipment and waste.
Medicinal extraction	Qualified dried biomass → manufacturer's approved extraction and purification → in-process tests → validated formulation	Licensed partner, approved process and product dossier. No generic farm recipe can substitute.
Chitin and chitosan	Qualified shell/residue → demineralization/deproteinization → deacetylation as needed → washing → drying → specifications	Chemical safety, wastewater treatment, molecular-weight/purity acceptance and patent review.
Frass	Separation → validated inactivation → stabilization → analysis → permitted use	Legal fertilizer or soil-product route and no viable insects/eggs.

A 2017 laboratory study reported 12.17% chitin and 7.28% chitosan recovery on a dry-weight basis. A 2026 study reported 17.78% chitin and 11.56% chitosan, using only 9 g of cockroach sample; it also disclosed a patent. Different inputs and methods prevent a simple industrial yield assumption. [S29] [S28]

Even an illustrative 10% chitosan yield from 300 kg dried biomass gives only 30 kg polymer per tonne fresh, before further qualification losses. Count chemical, water, effluent and quality costs, and deduct the biomass diverted from whole-insect sales. It is usually more credible to sell an approved residue to a specialist than to build a chemical plant at startup.

At the large 20 t fresh/month case, approximately 13.4 t of water must be removed monthly under the simple mass balance. At 22 processing days, that is about 609 kg/day. Quote drying capacity at the actual water-removal duty, peak loads and ambient winter/summer conditions, not merely kilograms of wet input.

Risk register and response plans

Risk and priority	Leading indicator	Mitigation and trigger
No legal route - critical	No written species/product determination	Stop procurement and product launch until resolved.
Buyer concentration - high	One buyer above 35% of sales	Diversify; secure minimum orders and payment protection; stress-test loss of largest buyer.
Price collapse - high	Net price below break-even for two months	Reduce cohort starts, remove uneconomic delivery routes, defer expansion.
Heat or power failure - critical	Alarm, temperature drift, failed backup test	Independent sensors, backup energy and isolated breeder reserve; rehearse response.
Escape - critical	Live insect/egg beyond containment zone	Stop movements, isolate room, investigate and follow local reporting and pest-response obligations.
Outbreak or unexplained mortality - high	Deaths double cohort baseline or cluster	Hold sales; isolate lots; obtain diagnostics; validate cleaning before restart.
Worker allergy - high	Respiratory or skin symptoms, visible dust	Exposure controls, health assessment and task redesign; never rely only on masks.
Contaminated feed or finished lot - critical	Positive tests, supplier change, off-odor	Quarantine, trace, test and recall if necessary; do not blend away a failure.
Working-capital shortfall - high	Cash below three-month fixed-cost floor	Freeze discretionary capital, tighten credit, secure funds before continued expansion.
Unproven processing or patent issue - high	Yield below target; customer rejects	Use toll trials and IP review before purchasing a full line.

These priorities are qualitative judgments, not actuarial probabilities. Assign each risk to a named owner, record incidents and review monthly. Insurance discussions should include product liability, public liability, equipment failure and business interruption, with explicit disclosure of the species and activity.

Do not use breeder-sale revenue or a vendor’s guaranteed buyback promise as proof of durable end-product demand. Verify the ultimate buyer, acceptance criteria, reject rights, payment history and enforceability before investing.

Sales strategy and customer validation

Build an addressable market from actual orders

Interview at least 20 prospective customers: specialist shops, reptile breeders, distributors, institutions and pharmaceutical procurement teams. Record current species, grade, monthly quantity, delivered price, seasonal demand, supplier problems, documentation and payment terms. A positive interview is a lead; a paid repeat order is evidence.

Illustrative monthly demand bridge	Small	Medium	Large
Consumer subscribers	100 × 100 = 10,000	500 × 100 = 50,000	Not assumed
Specialist stores	10 × 1,000 = 10,000	50 × 2,000 = 100,000	200 × 5,000 = 1m
Professional breeders/distributors	Not assumed	10 × 5,000 = 50,000	20 × 50,000 = 1m
Total target	20,000	200,000	2m

The table is a sales acquisition target, not a statement that these customers exist or will buy. Confirm that the reachable geography, shipping economics and species rules support the target. Avoid concentrating volume in low-margin distant orders.

Price around service economics

Offer weekly or fortnightly replenishment, predictable size grades, transparent counts and clear arrival terms. Set minimum order values by packaging and dispatch labor. For example, if packaging and unrecovered freight cost $4 per order, spreading that over 100 insects consumes $0.04 each before production costs. Consolidation or customer-paid freight is essential at low unit prices.

Track realized price after discounts, refunds, losses and marketplace fees; customer acquisition cost; repeat purchase; average order contribution; and complaints. Separate retained breeders from sellable inventory so marketing commitments do not reduce future production.

Contracting industrial or medicinal buyers

Require a technical specification, paid sample acceptance, forecast, minimum purchase, agreed assay and sampling method, delivery form, price adjustment, rejection process, payment term and change-control rules. For a large plant, seek commitments covering at least 70% of demonstrated capacity and test the remaining sales at conservative prices. A nonbinding letter of interest is insufficient to fund construction.

Implementation milestones and stop gates

Period after project start	Deliverable	Gate before next spending
Weeks 1-4	Confirm country/city, species law, product classification and 20 buyer interviews	Two credible product channels; documented regulatory route.
Months 1-3	Quarantine stock, compare feeds, instrument insulated trial room	Species identity, repeatable growth, acceptable contamination and containment.
Months 3-6	Paid size-graded sales and pilot processing samples	Positive contribution per order; documented buyer acceptance.
Months 6-12	Observe breeder replacement and winter operation	Measured energy, whole-farm feed cost, mortality and saleable output.
Months 12-18	Small commercial operation or medium modular expansion	At least three consecutive months meeting target economics and cash floor.
Months 18-30	Qualified raw-material line if justified	Three accepted production lots, legal status, paid contracts and engineering budget.
Month 24 onward	Consider large farm or industrial biomass modules	Reliable supply, buyer diversification, independent technical and financial review.

These are planning windows; species development, winter timing and approvals may extend them. Do not promise a full reproductive cycle inside an arbitrary calendar milestone. If the first pilot misses winter, postpone the large heating commitment until winter performance is known.

Documents required for an investment decision

Complete a site permit matrix; three equipment and construction quotations; delivered feed specifications and prices; a laboratory testing proposal; verified stock records; actual utility tariff and winter heat-load design; a cohort production model; customer contracts; a 24-month monthly cash forecast; and a contingency/recall plan.

Final go or no-go test

Proceed only when legally saleable output earns a positive contribution after all delivery costs, covers labor and winter energy at conservative utilization, and supports a funded cash reserve. Expand to medicinal materials only under a qualified buyer’s system. If feed-only customers cannot pay the full dried-product cost, compare an approved alternative insect species or stop that product route.

For the present site description, the recommended commitment is a $5,000-$10,000 validation program followed by a contained small farm only if its local sales economics work. Medium and large investments remain conditional.

Photographs and visual source notes



Photograph P5. Molting B. dubia, female left and male right, Colony BLABERUS, Czech Republic, 4 March 2016. Kolonie BLABERUS, CC BY-SA 4.0. Pale freshly molted individuals should not automatically be classified as diseased.

The five photographic plates show specimens, a captive colony, an egg capsule and molting. They are real photographs, not generated farm renderings. Photographs are reproduced proportionally without substantive alteration. No image depicts the proposed site. The climate chart and process flow are analytical graphics prepared for this plan.

P1 Drägüs. B. dubia colony at Schwerin Zoo and source file. CC BY-SA 3.0.

P2 Nicky.Animals. Male, female and juvenile B. dubia and source file. Public domain.

P3 Eugene Zelenko. P. americana specimen and source file. CC BY-SA 4.0.

P4 Eran Finkle. P. americana egg capsule and source file. CC BY-SA 3.0.

P5 Kolonie BLABERUS. Molting B. dubia and source file. CC BY-SA 4.0.

For photographs of actual Chinese industrial operations, consult the linked Sichuan Online 2025 facility article and Good Doctor’s 2025 site report. Their images are not reproduced here; retain the publisher’s rights and obtain permission before commercial reuse. [S13] [S15]

Sources and verification notes

Source links were checked during preparation on 8 November 2026. Dates below are publication dates where known. Seller prices can change; legal status must be reconfirmed for the actual site and intended use.

S01 Oonincx et al. (2015), Feed conversion, survival, development and composition of four insect species
Primary experiment; six replicate containers per diet. FCR and composition are laboratory results, not farm guarantees.

S02 Oonincx et al. (2019), Correction to the 2015 study
Corrects fatty-acid interpretation and associated tables. Avoid using uncorrected fatty-acid profiles.

S03 Wu, Hu and Appel (2017), Temperature-dependent development of Blaptica dubia
Primary thermal-development study; Journal of Economic Entomology 110, 546-551.

S04 Lin et al. (2022), Life-history traits of the American cockroach
Primary study at 29 C and 70 +/- 5% RH. DOI 10.3390/insects13060551.

S05 UF/IFAS, American cockroach species account
University extension reference; field and historical life-history values vary from controlled breeding trials.

S06 Batista et al. (2026), Dubia cockroaches fed commercial poultry feed
University repository abstract; DOI 10.1007/s42690-026-01958-5. Publication 18 July 2026.

S07 Mustafa et al. (2024), Periplaneta americana as a soybean meal substitute
Primary broiler feeding experiment; DOI 10.3390/insects15090632. Experimental substrate is not a legal commercial feed recommendation.

S08 Utsumi, Negishi and Kamei (2014), Effect of three rearing diets on cockroach longevity, fecundity and development
Primary feeding comparison; rat chow, beef and vegetables.

S09 TopFlight Dubia, adult breeding colony listings
Seller asking prices observed 8 November 2026; US retail, excluding unverified freight and taxes.

S10 Telos Exotics, live Dubia pricing
Seller asking prices observed 8 November 2026; US local retail, not East Asian farm-gate prices.

S11 Xinhua (29 August 2018), Cockroach factory treating kitchen waste
Historical facility report; reported waste intake and population are not audited output or profits.

Sources and verification notes

Source links were checked during preparation on 8 November 2026. Dates below are publication dates where known. Seller prices can change; legal status must be reconfirmed for the actual site and intended use.

S12 SCMP (19 April 2018), Giant indoor cockroach farm
Historical report of six billion adults per year; not a current global market estimate.

S13 Sichuan Online (15 April 2025), Digital cockroach breeding and pharmaceutical factory
Facility report; CNY 800 million refers to a wider pharmaceutical manufacturing project, not a standalone farm.

S14 Sichuan Economic Daily (26 February 2026), Good Doctor industrial transformation
Reports 22 billion cockroaches; a reported population figure, not verified annual saleable tonnage.

S15 Good Doctor Group (20 October 2025), University visit to Panxi facility
Company primary source confirming breeding, pharmaceutical and research activities; no open procurement commitment.

S16 JMA ClimatView, Beijing monthly climate normals
Official station data, 39.93 N; climate averages are not heating-system design extremes.

S17 JMA ClimatView, Akita monthly climate normals
Official station data, 39.72 N. Used to demonstrate variation at the same latitude.

S18 China government (2025), Implementation plan for feed-grain savings
Policy supports exploration of insect protein and catalogue development; does not approve every insect species.

S19 China joint agencies (2022), Good Agricultural Practice for traditional Chinese medicinal materials
Primary legal text hosted by FAO; covers medicinal plant and animal raw materials and origin processing.

S20 Japan MAFF, Feed Safety Act questions and answers
Official feed-safety framework; species, use, establishment and ingredient status require local confirmation.

S21 Republic of Korea MFDS (September 2024), Food-material guidance
Official publication lists ten edible insects at that date; cockroaches are not listed. Recheck current food code.

S22 EU Regulation 2025/1377
Official amendment; imported insect processed-animal-protein species list excludes cockroaches.

Sources and verification notes

Source links were checked during preparation on 8 November 2026. Dates below are publication dates where known. Seller prices can change; legal status must be reconfirmed for the actual site and intended use.

S23 FDA, Animal food ingredients
Official intended-use-specific ingredient requirements; live insect trading is not blanket ingredient approval.

S24 Australia DCCEEW, Live Import List
Updated 4 November 2026; explicit prohibition on possession, trade and breeding of Dubia and Madagascar hissing cockroaches.

S25 EFSA (2015), Risk profile of insects as food and feed
Official EFSA opinion, DOI 10.2903/j.efsa.2015.4257, mirrored by Feedipedia.

S26 Zeng et al. (2019), Periplaneta americana in modern and traditional Chinese medicine
Peer-reviewed review; describes Kangfuxin and Xinmailong. Product use does not validate a new extract.

S27 Lin et al. (2025), Kangfuxin in orthodontic childhood gingivitis
Systematic review of 19 studies; authors identify evidence-quality limitations. Not a basis for farm-level medical claims.

S28 Aderogba et al. (2026), Chitosan from prawn and American cockroach
Primary laboratory extraction; tiny sample, patent disclosed; yield is not an industrial process guarantee.

S29 Kim et al. (2017), Chitin and chitosan from cockroach exoskeleton
Primary study abstract; reports 12.17% chitin and 7.28% chitosan on dry weight.

S30 Innovafeed (15 July 2024), Third expansion phase at Nesle
Company source on black-soldier-fly industrial development; contextual competitor, not cockroach capacity.

S31 Le Monde (4 October 2026), French industrial insect farms
Contemporary reporting on restructuring, Ynsect bankruptcy and heat integration; not cockroach market sizing.

S32 FDA, Hazard analysis and preventive controls for animal food
Official technical guidance useful for process design; local compliance may differ.

S33 Japan JICA, Laws for edible and feed insects
Institutional regulatory overview; background only, not product approval.

S34 Li et al. (2026), Periplaneta americana as a protein feed ingredient in pigs
Primary study, Animal Feed Science and Technology 343, 116892. Abstract reports different performance for full-fat and unfatted material; not regulatory approval.